

SIDO KANHU MURMU UNIVERSITY



FOUR YEARS UNDER GRADUATE MINOR VOCATIONAL FOR B.A/B.SC/B.COM SYLLABUS

SUBJECT NAME- COMPUTER SCIENCE & IT

(UPDATED)

(As per New Education policy 2020)

Implementation from 2023-27

[Signature]
14.08.2023

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14-8-23

Co-Ordinator
Self Finance Course's
S.K.M.U. Dumka

Course Credits	No. of Teaching Hours Per Week	Total No. of Teaching Hours
04	04 Hrs	60 Hrs

Course Objective: To provide a basic understanding of computer and functionalities. The course aim to teach students about basic computer, operating system, internet, file management and cyber security.

Course Outcomes: This fundamental course outcomes provide students with a solid foundation in computer science and Technology and prepare them for more advanced coursed in this field.

UNIT – I Introduction to Computers

Computer system: characteristics and capabilities. Hardware and Software: Block Diagram of a Computer, Data Processing: Data, Data Processing System, Storing Data, Processing Data Types, Purpose, Components, Generation of Computers.

Unit-II Introduction to MS DOS & Windows

Fundamentals: Physical Structure of the Disk, Booting Process, Internal DOS Files & Directories, DOS Commands, Creating a Batch Files. Microsoft Windows: Operating system-Definition & functions, components.

UNIT – III File processing

word processing, creating and formatting , editing, creating table, graphic in documents, spreadsheet, formatting , formulas and function, data chart, sorting and filtering, introduction to presentation, editing, slideshow, drawing, creating presentation.

UNIT – IV Use of Computers in Education and Research:

Data analysis, heterogeneous storage, e-Library, Google scholar, Domain specific packages such as SPSS, Mathematica etc.

UNIT –V e-Governance

Meaning, origin, importance, objective, Fundamental, advantages Disadvantages, Challenges of e-Governance, Elements, Government to Citizens (G2C), Government to Business (G2B), Government to Government (G2G), Government to Employees (G2E)

Recommended Books:

1. Computer Fundamentals, P. K. Sinha, BPB Publications, Sixth Edition.
2. Introduction to Information Technology, V. Rajaraman, PHI, Second Edition.
3. Fundamental of Information Technology, ChetanShrivastava, Kalyani Publishers.

PAPER NAME: ARTIFICIAL INTELLIGENCE

FULL MARKS: 100 (External-75 : Internal-25)

DURATION: 3 Hrs.

Course Credits	No. of Teaching Hours Per Week	Total No. of Teaching Hours
04	04 Hrs	60 Hrs

Course Objective: The objective of Artificial Intelligence (AI) is to make machines particularly computers to do things that require intelligence when done by humans.

Course Outcomes: This AI course outcome provides students with the necessary knowledge, skills, and ethical awareness to become competent AI practitioners or researcher.

Unit I Introduction:

Intelligent Agents, Structure, environments, behaviour, nature of environments, Problem Solving agents. Example problems, searching for solutions, uniformed search strategies, avoiding repeated states.

Unit II Searching Technique:

Informed search and exploration, search strategies, heuristic function, local search algorithms and optimistic problems, local search in continuous spaces, online search agents and unknown environments.

Unit III Knowledge Representation:

First order logic , Syntax and semantics for first order logic , Knowledge engineering in first order logic - Inference in First order logic prepositional versus first order logic – unification and lifting – forward chaining – backward chaining.

Unit IV Learning:

Learning from observations - forms of learning - Inductive learning - Learning decision trees - Ensemble learning Logical formulation of learning – Explanation based learning – Learning using relevant information – Inductive logic programming.

Unit V Applications: Communication as action, Formal grammar for a fragment of English, Syntactic analysis, Augmented grammars, Semantic interpretation, Ambiguity and disambiguation, Discourse understanding, Grammar induction, Probabilistic language processing.

Recommended Books:

1. Ritch & Knight -Artificial Intelligence, TMH Reference Books:
1. S. Russel and P. Norvig- Artificial Intelligence A Modern Approach, Pearson Education.
2. Patterson -Introduction to Artificial Intelligence & Expert Systems, PHI

B.A/B.SC/B.COM: SIX SEMESTER

PAPER CODE: MN-VOC-66 2C

PAPER NAME: WEB DESIGNING

FULL MARKS: 100 (External-75 : Internal-25)

Pass marks - 40

DURATION: 3 Hrs.

Course Credits	No. of Teaching Hours Per Week	Total No. of Teaching Hours
04	04 Hrs	60 Hrs

Course Objective: The course aims at introducing the basic concepts and techniques of client side web programming. The student shall be able to develop simple websites using HTML, CSS and JavaScript.

Course Outcomes: Understanding of the principles of creating an effective web pages using PHP and Java script.

Unit I

HTML, Browsers, URL's, Domain Names, static and dynamic sites and active web pages, Files Creation, Web Server, HTML Tags, HTML Commands, External

Unit-II: Cascading Style Sheets (CSS)

Introduction to CSS, Features and benefits, syntax and structure, linking CSS, selectors, background images, colors and properties, manipulating texts, fonts, borders and boxes, margins, padding lists.

Unit III Introduction to JavaScript

JavaScript, Data Types and Literal, Type Casting, Variables, Array, Operators and Expressions, JavaScript Programming Constructs, If - then - else, Immediate If, For Loop, Built-in Functions, User Defined functions.

Unit IV Introduction to Internet

Concept of WWW, internet and WWW, protocols and programs, secure connections, application and development tools, web browser, URL, web server, web site designing principles, Domain Name.

Unit-V : Introduction to PHP

Introduction and basic syntax , Installing , Variables, Data Types, Strings, Constants, Operators, Programming Loops, Functions, Arrays, Forms and Form Handling, Form Validation PHP.

Recommended Books:

1. Robin Nixon, Learning PHP, MYSQL, JavaScript, CSS & HTML5 3ed: A Step-by-Step Guide to Creating Dynamic Websites, O'Reilly
2. Jon Duckett, HTML and CSS: Design and Build Websites, Wiley.
3. Jon Duckett, JavaScript and JQuery: Interactive Front-End Web Development, Wiley

B.A/B.SC/B.COM: EIGHT SEMESTER

PAPER CODE: MN-VOC-8D 2D

PAPER NAME: PROJECT WORK

FULL MARKS: 100 (External-75 : Internal-25)

PM-40

DURATION: 3 Hrs.

Course Credits	No. of Teaching Hours Per Week	Total No. of Teaching Hours
04	04 Hrs	60 Hrs

Project Component:

1. Design a home page which displays information about your college department using headings, HTML entities and paragraphs.
2. Implement different types of list tags, hyperlinks, marquee tag and HTML formatting tags in the college department homepage.
3. Create a web page having two frames, Frame 1 containing links and another with contents of the link. When a link is clicked appropriate contents should be displayed on Frame 2. Also, insert an iframe in the same page.
4. Design your course timetable and display it in tabular format.
5. Design an admission form for any course in your college with text, password fields, drop-down list, check-boxes, and radio buttons, submit and reset button etc. with proper CSS formatting.
6. Create a website for online book stores with Home, Login, Catalogue, Registration page with links to all these pages in a menu on top of every page. Embed heading, paragraph, images, video, .iframe, form controls, table, and list in this website. Use both Internal and external CSS in this.
7. Write a JavaScript program to display the current day and time.
8. Write a JavaScript program to a) Remove a character at the specified position of a given string and return the new string. b) Change the case of a string. (I.e. upper case to lower case and vice-versa).
9. Write a JavaScript program to compute the sum of elements of a given array of integers.
10. Develop and demonstrate a HTML file that includes JavaScript script for taking full name in a text field and display first, middle, last name *in 3 different labels. Middle and last name may be optional, thus messages like "NA" should be displayed in corresponding labels. If input contains 2 words, then they should be considered as first and last names. Page 28
11. Design HTML form for keeping student record, apply JavaScript validation for restriction of mandatory fields, numeric field, email-address field, specific value in a field etc.
12. Write a PHP script that removes the whitespaces from a string.